

Intermediate DSA Coding Practice — Questions #501–#525

Database-oriented problem specifications for Coding Hub ingestion. Each entry includes metadata, statement, constraints, I/O format, explanation, visible tests, and hidden tests.

#501. Longest Subarray With At Most K Distinct Values

Difficulty	Intermediate
Category	Arrays / Sliding Window
Slug	longest-subarray-at-most-k-distinct-values
Tags	array, sliding-window, hashmap, two-pointers

Problem Statement

Given an integer array and an integer K, find the length of the longest contiguous subarray containing at most K distinct values.

Constraints

$1 \leq N \leq 2 \times 10^5$; $1 \leq K \leq N$; $-10^9 \leq A[i] \leq 10^9$.

Input Format

First line: N K. Second line: N integers.

Output Format

Print the maximum valid subarray length.

Approach / Explanation

Maintain a frequency map for a moving window. Expand the right boundary and move the left boundary while the number of distinct values exceeds K.

Visible Test Cases

#	Input	Expected Output
1	7 2 1 2 1 2 3 2 2	5
2	5 1 4 4 4 2 4	3
3	6 3 1 2 3 4 5 6	3

Hidden Test Cases

#	Hidden Input	Expected Output
1	8 2 1 2 1 3 4 3 3 2	4
2	10 3 5 5 6 7 5 6 7 8 8 8	7

#502. Minimum Platforms Required for Train Schedule

Difficulty	Intermediate
Category	Arrays / Sorting / Greedy
Slug	minimum-platforms-train-schedule
Tags	sorting, greedy, intervals, two-pointers

Problem Statement

Given arrival and departure times of trains at a station, determine the minimum number of platforms required so that no train waits due to platform unavailability. A train arriving at the exact departure time of another train requires a separate platform.

Constraints

$1 \leq N \leq 10^5$; times are integers in HHMM format; every departure is after its corresponding arrival.

Input Format

First line: N. Second line: N arrival times. Third line: N departure times.

Output Format

Print the minimum required platforms.

Approach / Explanation

Sort arrivals and departures separately. Sweep through both arrays, increasing the active count for an arrival before or at the next departure, otherwise releasing a platform.

Visible Test Cases

#	Input	Expected Output
1	6 900 940 950 1100 1500 1800 910 1200 1120 1130 1900 2000	3
2	3 100 200 300 150 250 350	1
3	4 900 900 900 900 1000 1000 1000 1000	4

Hidden Test Cases

#	Hidden Input	Expected Output
1	5 900 901 902 903 904 905 906 907 908 909	5
2	5 1000 1015 1030 1045 1100 1010 1020 1040 1050 1110	2

#503. Count Subarrays With Sum Divisible by K

Difficulty	Intermediate
Category	Arrays / Prefix Sum
Slug	count-subarrays-sum-divisible-by-k
Tags	prefix-sum, hashmap, modular-arithmetic

Problem Statement

Count the number of contiguous subarrays whose sum is divisible by K. The array may contain negative values.

Constraints

$1 \leq N \leq 2 \times 10^5$; $-10^4 \leq A[i] \leq 10^4$; $1 \leq K \leq 10^5$.

Input Format

First line: N K. Second line: N integers.

Output Format

Print the number of qualifying subarrays.

Approach / Explanation

Two prefix sums with the same normalized remainder modulo K define a subarray whose sum is divisible by K. Store frequencies of remainders.

Visible Test Cases

#	Input	Expected Output
1	6 5 4 5 0 -2 -3 1	7
2	3 3 1 2 3	3
3	5 2 1 1 1 1 1	6

Hidden Test Cases

#	Hidden Input	Expected Output
1	4 7 7 -7 14 -14	10
2	6 4 -1 2 3 -4 5 -5	6

#504. Merge Overlapping Intervals With Minimum Output

Difficulty	Intermediate
Category	Intervals / Sorting
Slug	merge-overlapping-intervals
Tags	intervals, sorting, merging

Problem Statement

Given N closed intervals, merge every pair of overlapping or touching intervals and output the resulting disjoint intervals in ascending order.

Constraints

$1 \leq N \leq 10^5$; $0 \leq \text{start} \leq \text{end} \leq 10^9$.

Input Format

First line: N . Next N lines: start end.

Output Format

First print the number of merged intervals, followed by each merged interval on a separate line.

Approach / Explanation

Sort intervals by starting point. Extend the current interval whenever the next interval begins no later than $\text{current_end} + 1$; otherwise finalize the current interval.

Visible Test Cases

#	Input	Expected Output
1	4 1 3 2 6 8 10 9 12	2 1 6 8 12
2	3 1 2 3 4 5 6	1 1 6
3	3 5 7 1 2 10 12	3 1 2 5 7 10 12

Hidden Test Cases

#	Hidden Input	Expected Output
1	5 1 10 2 3 4 8 11 12 12 15	2 1 10 11 15
2	4 0 0 0 2 3 3 5 9	2 0 3 5 9

#505. Kth Smallest Element in a Sorted Matrix

Difficulty	Intermediate
Category	Binary Search / Matrix
Slug	kth-smallest-sorted-matrix
Tags	binary-search, matrix, counting

Problem Statement

You are given an $N \times N$ matrix where every row and every column is sorted in nondecreasing order. Find the Kth smallest element.

Constraints

$1 \leq N \leq 500$; $1 \leq K \leq N^2$; matrix values lie between -10^9 and 10^9 .

Input Format

First line: N K. Next N lines contain N integers each.

Output Format

Print the Kth smallest element.

Approach / Explanation

Binary-search the answer value. For each candidate, count how many matrix elements are less than or equal to it using a staircase traversal.

Visible Test Cases

#	Input	Expected Output
1	3 8 1 5 9 10 11 13 12 13 15	13
2	2 2 1 2 3 4	2
3	3 1 -5 -4 -1 0 2 3 4 8 9	-5

Hidden Test Cases

#	Hidden Input	Expected Output
1	4 10 1 2 4 8 3 5 7 9 6 10 12 14 11 13 15 20	10
2	2 4 -3 -1 0 6	6

#506. Shortest Path in a Binary Matrix

Difficulty	Intermediate
Category	Graphs / BFS
Slug	shortest-path-binary-matrix
Tags	graph, bfs, grid, shortest-path

Problem Statement

Given a square binary grid, move in any of the eight directions from a cell containing 0 to another cell containing 0. Find the length of the shortest path from the top-left cell to the bottom-right cell, counting both endpoints.

Constraints

$1 \leq N \leq 1000$; grid values are 0 or 1.

Input Format

First line: N. Next N lines contain N binary values.

Output Format

Print the shortest path length, or -1 if unreachable.

Approach / Explanation

Use BFS because every move has equal cost. Mark visited cells when enqueueing them to avoid repeated processing.

Visible Test Cases

#	Input	Expected Output
1	3 0 0 0 1 1 0 1 1 0	4
2	2 0 1 1 0	-1
3	3 0 0 0 0 1 0 0 0 0	3

Hidden Test Cases

#	Hidden Input	Expected Output
1	4 0 1 0 0 0 1 0 1 0 0 0 1 1 1 0 0	5
2	1 0	1

#507. Course Schedule Feasibility

Difficulty	Intermediate
Category	Graphs / Topological Sort
Slug	course-schedule-feasibility
Tags	graph, dag, topological-sort, cycle-detection

Problem Statement

There are N courses numbered from 0 to $N-1$ and prerequisite pairs $[a,b]$, meaning course b must be completed before course a . Determine whether all courses can be completed.

Constraints

$1 \leq N \leq 10^5$; $0 \leq M \leq 2 \times 10^5$; course IDs are valid.

Input Format

First line: N M . Next M lines: a b .

Output Format

Print YES if all courses can be completed; otherwise print NO.

Approach / Explanation

Build a directed graph and compute indegrees. Kahn's algorithm can process every node exactly when the graph is acyclic.

Visible Test Cases

#	Input	Expected Output
1	2 1 1 0	YES
2	2 2 1 0 0 1	NO
3	4 4 1 0 2 1 3 2 3 0	YES

Hidden Test Cases

#	Hidden Input	Expected Output
1	5 5 1 0 2 0 3 1 3 2 4 3	YES
2	3 3 0 1 1 2 2 0	NO

#508. Network Delay Time

Difficulty	Intermediate
Category	Graphs / Dijkstra
Slug	network-delay-time
Tags	weighted-graph, dijkstra, shortest-path

Problem Statement

A directed weighted network contains N nodes. A signal starts at node K and travels along directed edges. Find the time needed for all nodes to receive the signal, or -1 if some node is unreachable.

Constraints

$1 \leq N \leq 10^5$; $0 \leq M \leq 2 \times 10^5$; edge weights are positive and at most 10^6 .

Input Format

First line: N M K . Next M lines: u v w .

Output Format

Print the maximum shortest-path distance from K to any node, or -1 .

Approach / Explanation

Use Dijkstra's algorithm with a min-priority queue. The answer is the largest finalized shortest distance.

Visible Test Cases

#	Input	Expected Output
1	4 4 2 2 1 1 2 3 1 3 4 1 1 4 5	2
2	3 1 1 1 2 4	-1
3	5 6 1 1 2 2 1 3 5 2 3 1 3 4 2 4 5 3 2 5 10	8

Hidden Test Cases

#	Hidden Input	Expected Output
1	4 5 1 1 2 10 1 3 3 3 2 2 2 4 1 3 4 8	6
2	2 1 2 1 2 7	-1

#509. Longest Increasing Subsequence With Reconstruction

Difficulty	Intermediate
Category	Dynamic Programming / Binary Search
Slug	longest-increasing-subsequence-reconstruction
Tags	dp, lis, binary-search, reconstruction

Problem Statement

Find one strictly increasing subsequence of maximum length from an array. Output its length and the elements of one valid subsequence.

Constraints

$1 \leq N \leq 2 \times 10^5$; $-10^9 \leq A[i] \leq 10^9$.

Input Format

First line: N. Second line: N integers.

Output Format

First print the LIS length. On the next line print one LIS in order.

Approach / Explanation

Maintain the smallest possible tail value for each subsequence length and predecessor indices to reconstruct one optimal subsequence.

Visible Test Cases

#	Input	Expected Output
1	8 10 9 2 5 3 7 101 18	4 2 3 7 101
2	5 5 4 3 2 1	1 5
3	6 1 2 2 3 4 1	4 1 2 3 4

Hidden Test Cases

#	Hidden Input	Expected Output
1	7 7 1 5 2 6 3 9	4 1 2 6 9
2	4 -4 -2 -3 0	3 -4 -2 0

#510. Partition Array Into Equal-Sum Subsets

Difficulty	Intermediate
Category	Dynamic Programming / Bitmask
Slug	partition-array-equal-sum-subsets
Tags	dp, subset-sum, bitmask

Problem Statement

Given an array of positive integers and an integer K, determine whether the array can be partitioned into exactly K non-empty subsets such that every subset has the same sum.

Constraints

$1 \leq N \leq 16$; $1 \leq K \leq N$; $1 \leq A[i] \leq 10^8$.

Input Format

First line: N K. Second line: N integers.

Output Format

Print YES if such a partition exists; otherwise print NO.

Approach / Explanation

If total sum is not divisible by K, the answer is NO. Use subset-mask DP or backtracking with memoization to build groups of target sum.

Visible Test Cases

#	Input	Expected Output
1	7 4 4 3 2 3 5 2 1	YES
2	5 3 1 2 3 4 5	NO
3	6 2 2 2 2 2 3 3	YES

Hidden Test Cases

#	Hidden Input	Expected Output
1	8 4 1 1 1 1 2 2 2 2	YES
2	6 3 2 2 2 2 2 2	YES

#511. Edit Distance Between Two Strings

Difficulty	Intermediate
Category	Dynamic Programming / Strings
Slug	edit-distance-two-strings
Tags	dp, strings, levenshtein-distance

Problem Statement

Given two strings, calculate the minimum number of insertions, deletions, and substitutions required to transform the first string into the second.

Constraints

$0 \leq |A|, |B| \leq 2000$; strings contain lowercase English letters.

Input Format

Two lines containing strings A and B.

Output Format

Print the minimum edit distance.

Approach / Explanation

Let $dp[i][j]$ be the minimum operations to transform the first i characters into the first j characters of the second string.

Visible Test Cases

#	Input	Expected Output
1	horse ros	3
2	intention execution	5
3	abc abc	0

Hidden Test Cases

#	Hidden Input	Expected Output
1	kitten sitting	3
2	abc xyz	3

#512. Minimum Coins for Exact Amount

Difficulty	Intermediate
Category	Dynamic Programming / Unbounded Knapsack
Slug	minimum-coins-exact-amount
Tags	dp, coin-change, unbounded-knapsack

Problem Statement

Given unlimited supplies of coins and a target amount, find the minimum number of coins needed to form the exact amount. If impossible, print -1.

Constraints

$1 \leq N \leq 100$; $1 \leq \text{coin value} \leq 10^4$; $0 \leq \text{target} \leq 10^5$.

Input Format

First line: N target. Second line: N distinct positive coin values.

Output Format

Print the minimum number of coins, or -1.

Approach / Explanation

Use one-dimensional unbounded knapsack DP where $dp[x]$ is the minimum coins needed to form amount x .

Visible Test Cases

#	Input	Expected Output
1	3 11 1 2 5	3
2	2 3 2 4	-1
3	4 0 2 3 5 7	0

Hidden Test Cases

#	Hidden Input	Expected Output
1	3 27 2 5 10	5
2	5 63 1 7 10 21 25	3

#513. Maximum Product Subarray

Difficulty	Intermediate
Category	Arrays / Dynamic Programming
Slug	maximum-product-subarray
Tags	array, dp, greedy

Problem Statement

Find the maximum product obtainable from a non-empty contiguous subarray. The array may contain negative values and zeros.

Constraints

$1 \leq N \leq 2 \times 10^5$; $-10 \leq A[i] \leq 10$.

Input Format

First line: N. Second line: N integers.

Output Format

Print the maximum subarray product.

Approach / Explanation

Track both maximum and minimum products ending at the current position because multiplying by a negative number swaps their roles.

Visible Test Cases

#	Input	Expected Output
1	4 2 3 -2 4	6
2	3 -2 0 -1	0
3	3 -2 3 -4	24

Hidden Test Cases

#	Hidden Input	Expected Output
1	5 -1 -2 -3 0 -4	6
2	6 -2 -3 0 -2 -40 4	80

#514. Find All Anagram Starting Indices

Difficulty	Intermediate
Category	Strings / Sliding Window
Slug	find-all-anagram-starting-indices
Tags	strings, hashmap, sliding-window

Problem Statement

Given strings S and P, find every starting index in S where an anagram of P occurs as a contiguous substring. Indices are zero-based.

Constraints

$1 \leq |P| \leq |S| \leq 2 \times 10^5$; lowercase English letters only.

Input Format

First line: S. Second line: P.

Output Format

Print the number of indices followed by the indices in ascending order.

Approach / Explanation

Compare character frequencies in a fixed-size sliding window of length $|P|$ against the frequency vector of P.

Visible Test Cases

#	Input	Expected Output
1	cbaebabacd abc	2 0 6
2	abab ab	3 0 1 2
3	abcdef gh	0

Hidden Test Cases

#	Hidden Input	Expected Output
1	baa aa	1 1
2	abbcabc abc	2 2 4

#515. Decode Nested Repetition String

Difficulty	Intermediate
Category	Stacks / Strings
Slug	decode-nested-repetition-string
Tags	stack, parsing, strings

Problem Statement

Decode an encoded string where patterns follow the form $k[\text{encoded_text}]$. The encoded text may contain nested patterns. Return the fully decoded string.

Constraints

Encoded length ≤ 5000 ; repetition counts are between 1 and 10^4 ; decoded output length $\leq 10^6$.

Input Format

One line containing the encoded string.

Output Format

Print the decoded string.

Approach / Explanation

Use stacks for repetition counts and partial strings. When a closing bracket appears, pop the latest context and repeat the completed segment.

Visible Test Cases

#	Input	Expected Output
1	3[a]2[bc]	aaabcbc
2	3[a2[c]]	accaccacc
3	2[ab3[c]]	abcccabccc

Hidden Test Cases

#	Hidden Input	Expected Output
1	10[x]	xxxxxxxxxx
2	2[a2[b2[c]]]	abcbcbcbcbcbcbcb

#516. Largest Rectangle in Histogram

Difficulty	Intermediate
Category	Stacks / Arrays
Slug	largest-rectangle-histogram
Tags	monotonic-stack, histogram

Problem Statement

Given bar heights in a histogram where each bar has width one, find the area of the largest rectangle formed by consecutive bars.

Constraints

$1 \leq N \leq 2 \times 10^5$; $0 \leq \text{height} \leq 10^9$.

Input Format

First line: N. Second line: N heights.

Output Format

Print the maximum rectangle area.

Approach / Explanation

Use a monotonic increasing stack to determine the nearest smaller bar on both sides for each height.

Visible Test Cases

#	Input	Expected Output
1	6 2 1 5 6 2 3	10
2	2 2 2	4
3	5 1 2 3 4 5	9

Hidden Test Cases

#	Hidden Input	Expected Output
1	7 6 2 5 4 5 1 6	12
2	4 0 0 0 0	0

#517. Serialize and Deserialize a Binary Tree

Difficulty	Intermediate
Category	Trees / Traversal
Slug	serialize-deserialize-binary-tree
Tags	binary-tree, dfs, serialization

Problem Statement

Given a binary tree in level-order form, serialize it using preorder traversal with null markers, then deserialize the representation and print its inorder traversal. The goal is to preserve the exact tree structure.

Constraints

$1 \leq \text{number of nodes} \leq 10^4$; node values range from -10^9 to 10^9 .

Input Format

First line: N. Second line: level-order tokens where # represents null.

Output Format

Print the serialized preorder representation and the inorder traversal after deserialization.

Approach / Explanation

A preorder traversal with explicit null markers uniquely identifies a binary tree. Deserialize by consuming tokens recursively.

Visible Test Cases

#	Input	Expected Output
1	7 1 2 3 # # 4 5	1 2 # # 3 4 # # 5 # # 2 1 4 3 5
2	1 7	7 # # 7
3	3 1 # 2	1 # 2 # # 1 2

Hidden Test Cases

#	Hidden Input	Expected Output
1	7 10 5 15 2 7 12 20	10 5 2 # # 7 # # 15 12 # # 20 # # 2 5 7 10 12 15 20
2	1 -4	-4 # # -4

#518. Lowest Common Ancestor in a Binary Tree

Difficulty	Intermediate
Category	Trees / Recursion
Slug	lowest-common-ancestor-binary-tree
Tags	binary-tree, lca, recursion

Problem Statement

Given a binary tree and two node values guaranteed to exist in the tree, find their lowest common ancestor. Values are unique.

Constraints

$1 \leq N \leq 10^5$; node values are unique integers.

Input Format

First line: N. Second line: level-order representation with # for null. Third line: two target values.

Output Format

Print the value of the lowest common ancestor.

Approach / Explanation

Recursively search both subtrees. If one target is found in each subtree, the current node is the LCA; if the current node matches either target, return it upward.

Visible Test Cases

#	Input	Expected Output
1	7 3 5 1 6 2 0 8 5 1	3
2	5 3 5 1 6 2 6 2	5
3	3 1 2 3 2 3	1

Hidden Test Cases

#	Hidden Input	Expected Output
1	7 10 5 15 3 7 12 20 3 7	5
2	7 10 5 15 3 7 12 20 12 20	15

#519. Detect Cycle in an Undirected Graph

Difficulty	Intermediate
Category	Graphs / DFS-BFS
Slug	detect-cycle-undirected-graph
Tags	graph, cycle-detection, dfs, union-find

Problem Statement

Given an undirected graph with N vertices and M edges, determine whether the graph contains at least one cycle.

Constraints

$1 \leq N \leq 10^5$; $0 \leq M \leq 2 \times 10^5$; self-loops and parallel edges may appear.

Input Format

First line: N M . Next M lines: u v using 1-based vertices.

Output Format

Print YES if a cycle exists; otherwise print NO.

Approach / Explanation

Use DFS with parent tracking or Disjoint Set Union. An edge connecting two already-connected vertices indicates a cycle.

Visible Test Cases

#	Input	Expected Output
1	3 3 1 2 2 3 3 1	YES
2	4 3 1 2 2 3 3 4	NO
3	2 2 1 2 1 2	YES

Hidden Test Cases

#	Hidden Input	Expected Output
1	5 4 1 2 2 3 4 5 1 3	YES
2	1 1 1 1	YES

#520. Word Ladder Shortest Transformation

Difficulty	Intermediate
Category	Graphs / BFS / Strings
Slug	word-ladder-shortest-transformation
Tags	bfs, graph, strings, shortest-path

Problem Statement

Transform a begin word into an end word by changing exactly one character at a time. Every intermediate word must belong to a supplied dictionary. Return the number of words in the shortest valid transformation sequence, including both endpoints.

Constraints

$1 \leq \text{dictionary size} \leq 5000$; all words have equal length ≤ 10 ; lowercase letters only.

Input Format

First line: beginWord endWord. Second line: M. Next M lines: dictionary words.

Output Format

Print the shortest sequence length, or 0 if impossible.

Approach / Explanation

Treat words as graph nodes and connect words differing by one character. BFS finds the shortest transformation.

Visible Test Cases

#	Input	Expected Output
1	hit cog 6 hot dot dog lot log cog	5
2	hit cog 5 hot dot dog lot log	0
3	a c 3 a b c	2

Hidden Test Cases

#	Hidden Input	Expected Output
1	talk tail 5 talk tall tail balk bail	3
2	lost cost 4 most cost lost host	2

#521. House Robber on a Circular Street

Difficulty	Intermediate
Category	Dynamic Programming
Slug	house-robber-circular-street
Tags	dp, circular-array, optimization

Problem Statement

Houses are arranged in a circle. Adjacent houses cannot both be robbed because of an alarm. Find the maximum amount of money that can be robbed without robbing neighboring houses.

Constraints

$1 \leq N \leq 2 \times 10^5$; $0 \leq \text{money}[i] \leq 10^9$.

Input Format

First line: N. Second line: N nonnegative integers.

Output Format

Print the maximum obtainable amount.

Approach / Explanation

The first and last houses cannot both be selected. Solve two linear cases: exclude the first house or exclude the last house.

Visible Test Cases

#	Input	Expected Output
1	4 2 3 2 3	6
2	3 1 2 3	3
3	1 7	7

Hidden Test Cases

#	Hidden Input	Expected Output
1	5 5 1 1 5 2	10
2	6 10 2 2 10 2 2	20

#522. Target Sum Ways

Difficulty	Intermediate
Category	Dynamic Programming / Subset Sum
Slug	target-sum-ways
Tags	dp, subset-sum, counting

Problem Statement

Assign either + or – before every number in an array so that the resulting expression equals target. Count the number of valid assignments.

Constraints

$1 \leq N \leq 20$; $0 \leq A[i] \leq 1000$; $-10^4 \leq \text{target} \leq 10^4$.

Input Format

First line: N target. Second line: N integers.

Output Format

Print the number of valid sign assignments.

Approach / Explanation

Use memoized recursion over index and current sum, or transform the problem into counting subsets when applicable.

Visible Test Cases

#	Input	Expected Output
1	5 3 1 1 1 1 1	5
2	3 0 0 0 0	8
3	2 1 1 2	1

Hidden Test Cases

#	Hidden Input	Expected Output
1	4 2 1 2 3 4	2
2	6 0 1 1 1 1 1 1	20

#523. Minimum Window Containing All Characters

Difficulty	Intermediate
Category	Strings / Sliding Window
Slug	minimum-window-containing-all-characters
Tags	strings, sliding-window, hashmap

Problem Statement

Given strings S and T, find the shortest substring of S that contains every character of T with at least the required frequency. If no such substring exists, print -1.

Constraints

$1 \leq |T| \leq |S| \leq 2 \times 10^5$; strings contain upper- and lowercase English letters.

Input Format

First line: S. Second line: T.

Output Format

Print the shortest valid substring, or -1.

Approach / Explanation

Expand a right pointer until all required characters are satisfied, then shrink from the left while maintaining validity.

Visible Test Cases

#	Input	Expected Output
1	ADOBECODEBANC ABC	BANC
2	a a	a
3	a b	-1

Hidden Test Cases

#	Hidden Input	Expected Output
1	aaabdbabcefaecbef abc	abc
2	this is a test string tist	t stri

#524. Rotate a Linked List by K Positions

Difficulty	Intermediate
Category	Linked Lists
Slug	rotate-linked-list-k-positions
Tags	linked-list, pointers, modular-arithmetic

Problem Statement

Given a singly linked list, rotate it to the right by K positions and print the resulting list.

Constraints

$0 \leq N \leq 10^5$; $0 \leq K \leq 10^{18}$; node values range from -10^9 to 10^9 .

Input Format

First line: N K. Second line: N node values.

Output Format

Print the rotated list.

Approach / Explanation

Connect the tail to the head to form a cycle, then break the cycle immediately before the new head after reducing K modulo N.

Visible Test Cases

#	Input	Expected Output
1	5 2 1 2 3 4 5	4 5 1 2 3
2	3 0 1 2 3	1 2 3
3	1 99 7	7

Hidden Test Cases

#	Hidden Input	Expected Output
1	6 8 1 2 3 4 5 6	5 6 1 2 3 4
2	4 6 -1 -2 -3 -4	-3 -4 -1 -2

#525. Find Duplicate Number Without Extra Space

Difficulty	Intermediate
Category	Arrays / Cycle Detection
Slug	find-duplicate-number-without-extra-space
Tags	array, floyd-cycle-detection, pointers

Problem Statement

An array contains $N+1$ integers where each value lies between 1 and N . Exactly one value appears at least twice. Find the duplicated value without modifying the array and using $O(1)$ extra space.

Constraints

$1 \leq N \leq 10^5$; values are in $[1, N]$.

Input Format

First line: N . Second line: $N+1$ integers.

Output Format

Print the duplicate value.

Approach / Explanation

Interpret the array as a functional graph and apply Floyd's tortoise-and-hare cycle detection to find the cycle entrance.

Visible Test Cases

#	Input	Expected Output
1	4 1 3 4 2 2	2
2	3 1 3 3 2	3
3	5 2 5 1 1 4 3	1

Hidden Test Cases

#	Hidden Input	Expected Output
1	6 6 4 3 2 5 1 6	6
2	2 1 1 2	1